Status	Finished
Started	Wednesday, 21 October 2026, 5:04 PM
Completed	Wednesday, 21 October 2026, 5:07 PM
Duration	2 mins 54 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Sequential search
c. Binary search ✓
d. Hash search

Question **2**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Jump search
c. Linear search ✓
d. Binary search

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94
b. 90 and 99 ✓
c. 94 and 99
d. 90 and 100

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O👎
c. O(n log n)
d. O(log n) ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search starts over from the beginning
c. The search stops immediately ✓
d. The search continues until the end of the list

Question **6**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓
b. Hash search
c. Both (1) & (3)
d. Binary search

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The search is considered unsuccessful ✓
c. An error is raised
d. The last element is returned

Question **8**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. 1
c. n/2
d. n ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list contains duplicate elements
c. When the list is very small
d. When the list is sorted ✓

Question **10**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list
b. Sorting elements in a list
c. Determining whether an element is present in a list ✓
d. Deleting elements from a list

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search requires the list to be sorted.
c. Linear search divides the list into halves.
d. Linear search is more efficient than binary search on large lists.

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It does not work with negative numbers
d. It requires the list to be sorted ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓
b. Dividing the list in half repeatedly
c. Skipping every second element
d. Sorting the list before searching

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94
b. 89 and 94
c. 89 and 99
d. 90 and 99 ✓

Question **15**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 100
c. 500
d. 1000